MAE 155A: Aerospace Engineering Design I

Cost analysis

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Cost analysis overview

- The cost analysis considers the **full lifecycle**.
- We will make use of empirical **cost-estimating relationships (CERs)**.
- It is very important to consider the **time value of money**

$$NPV = \frac{FV}{(1+r)^n}$$

- When estimating labor hours, we must consider **learning curve effects**—production costs decrease with each aircraft produced

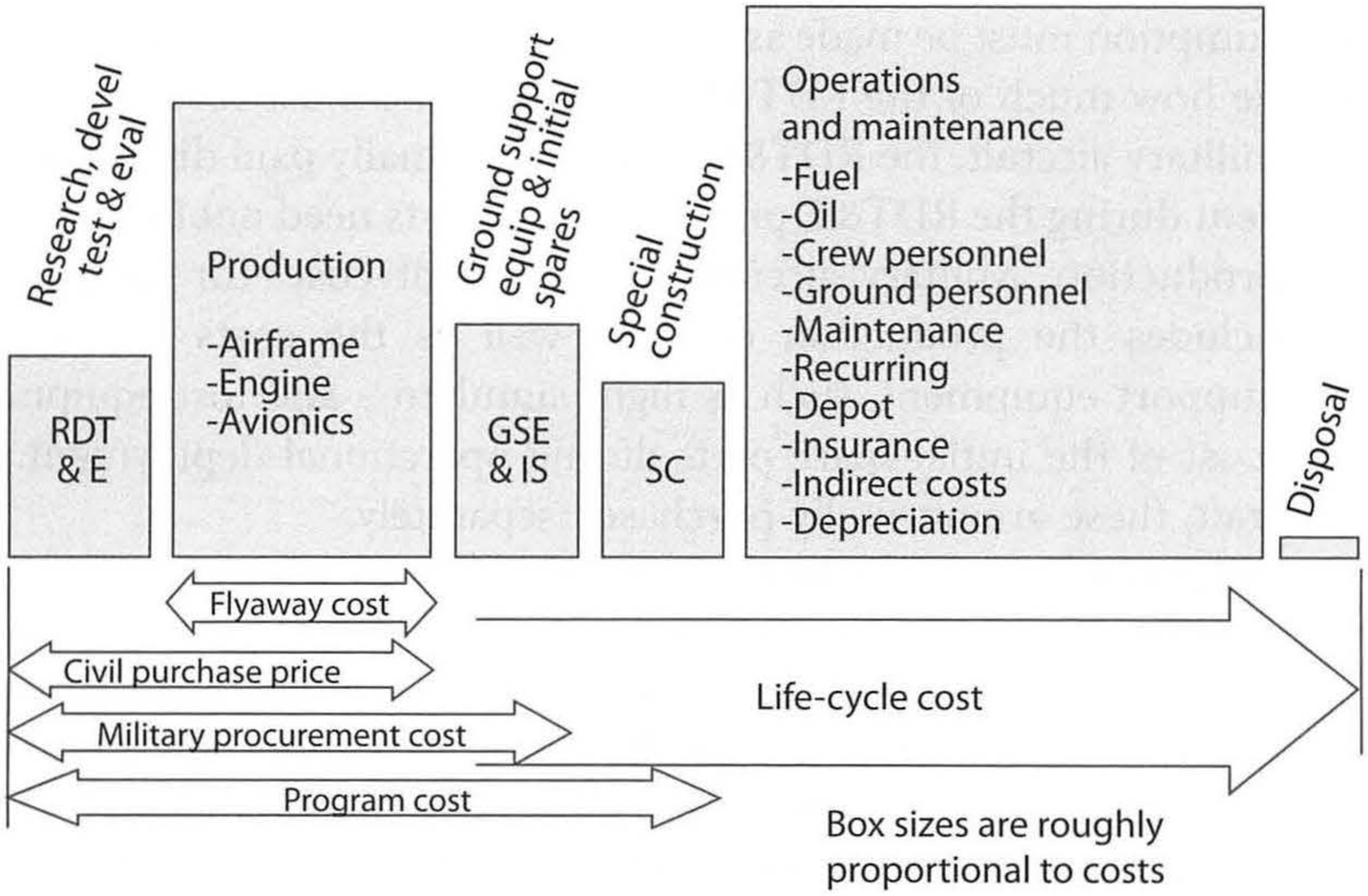
$$H = H_1 \left(\frac{1}{Q}\right)^{\log_2 \text{learning curve}}$$

Lifecycle cost

In aircraft produced in large volume, operations & maintenance dominates

Fuel / electricity and maintenance is often a good surrogate for lifecycle cost as an objective function

Drones: crew personnel would be replaced by support staff and remote operators



Raymer, D. (2012). Aircraft design: a conceptual approach. American Institute of Aeronautics and Astronautics, Inc.

Operating costs

- Total operating cost: TOC = DOC + IOC
  - Direct operating cost: DOC = FOC + COC
    - Fixed operating cost: FOC, e.g., depreciation
    - Cash operating cost: COC, e.g., incurred from operating flights

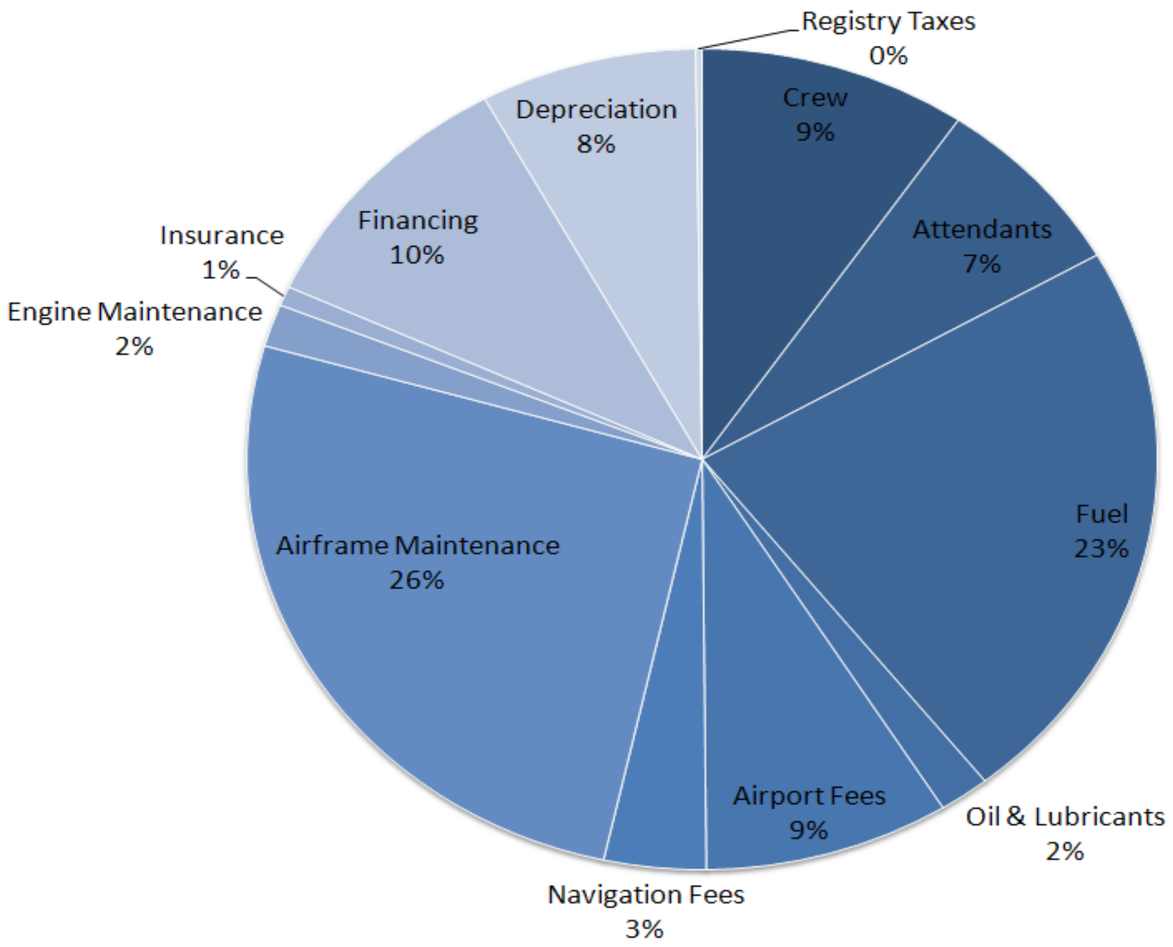
DOC:

- Costs associated with flight operations: fuel, oil, crew, maintenance, depreciation, insurance, airport fees, carbon taxes.
- DOC is typically in tens of cents per seat-mile for airliners. DOC is a few dollars per seat-mile for air taxis. Tens to low thousands of dollars per flight-hour for military aircraft.
- Good choice as an objective function in aircraft design.

IOC:

- General costs associated with airline operations: depreciation of ground facilities and equipment, sales and customer service costs, and administrative and overhead costs.
- Not strongly correlated to the design of the aircraft—varies across airlines. Typically 1/3 to 100% of DOC.

Direct operating cost (DOC)



Estimating crew and maintenance costs

Aircraft class	FH/YR/AC	Crew ratio	MMH/FH
Light aircraft	500–1000	—	1/4–1
Business jet	500–2000	—	3–6
Jet trainer	300–500	—	6–10
Fighter (modern)	300–500	1.1	10–15
Bomber	300–500	1.5	25–50
Military transport	700–1400	1.5 if FH/YR < 1200 2.5 if 1200 < FH/YR < 2400 3.5 if 2400 < FH/YR	20–40
Civil transport	2500–4500	—	5–15

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- FH/YR/AC = flight hours per year per aircraft
- Crew ratio = number of aircrews per aircraft
- MMH/FH = maintenance hours per flight hours

Direct operating cost (DOC)

Wrap rates: multiplier on labor wages to account for benefits, overhead, and administrative costs.

Depreciation: annual allocation of the purchase price over the aircraft life. If the resale value is 10% of the purchase price after n years, the contribution to operating cost from depreciation is 0.9 times the purchase price divided by n.

Insurance: 1–3% of the operating cost.

Maintenance: estimate based on MMH/FH and equations below (costs are in 2012 dollars).

$$\frac{\text{material cost}}{\text{FH}} = 3.3\left(\frac{C_a}{10^6}\right) + 14.2 + \left[58\left(\frac{C_e}{10^6}\right) - 26.1\right] N_e$$

$$\frac{\text{material cost}}{\text{cycle}} = 4.0\left(\frac{C_a}{10^6}\right) + 9.3 + \left[7.5\left(\frac{C_e}{10^6}\right) + 5.6\right] N_e$$

C_a = aircraft cost less engine

C_e = cost per engine

N_e = number of engines

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RDT&E and production costs

RDT&E: Research, Development, Testing, & Evaluation

- **Engineering hours:** analysis, design, testing (airframe and propulsion integration).
- **Tooling hours:** preparation for production—tools, fixtures, molds, test apparatus.
- **Manufacturing labor:** forming, machining, fastening, subassembly fabrication, final assembly, routing, part installation.
- **Quality control:** inspection of aircraft parts, tools, and fixtures.
- **Development support:** mockups, simulators, test articles.
- **Flight test:** testing for airworthiness and certification, but not the cost of the test aircraft themselves (covered by engineering production).
- **Manufacturing materials:** structural raw materials.
- **Engine production:** cost of producing the engine, not including the cost of developing a new one.

Note: Predicted costs should be multiplied by an investment cost factor of 1.1–1.4 to account for interest and contractor profit.

Cost-estimating relationships (CERs) from RAND Corp.

$$\text{Eng hours} = H_E = 4.86 W_e^{0.777} V^{0.894} Q^{0.163} \quad \{\text{fps}\}$$
$$= 5.18 W_e^{0.777} V^{0.894} Q^{0.163} \quad \{\text{mks}\}$$
$$\text{Tooling hours} = H_T = 5.99 W_e^{0.777} V^{0.696} Q^{0.263} \quad \{\text{fps}\}$$
$$= 7.22 W_e^{0.777} V^{0.696} Q^{0.263} \quad \{\text{mks}\}$$
$$\text{Mfg hours} = H_M = 7.37 W_e^{0.82} V^{0.484} Q^{0.641} \quad \{\text{fps}\}$$
$$= 10.5 W_e^{0.82} V^{0.484} Q^{0.641} \quad \{\text{mks}\}$$
$$\text{QC hours} = H_Q = 0.076 \text{ (mfg hours) if cargo airplane}$$
$$= 0.133 \text{ (mfg hours) otherwise}$$
$$\text{Devel support cost} = C_D = 91.3 W_e^{0.630} V^{1.3} \quad \{\text{fps}\}$$
$$= 67.4 W_e^{0.630} V^{1.3} \quad \{\text{mks}\}$$
$$\text{Flt test cost} = C_F = 2498 W_e^{0.325} V^{0.822} \text{FTA}^{1.21} \quad \{\text{fps}\}$$
$$= 1947 W_e^{0.325} V^{0.822} \text{FTA}^{1.21} \quad \{\text{mks}\}$$
$$\text{Mfg materials cost} = C_M = 22.1 W_e^{0.921} V^{0.621} Q^{0.799} \quad \{\text{fps}\}$$
$$= 31.2 W_e^{0.921} V^{0.621} Q^{0.799} \quad \{\text{mks}\}$$
$$\text{Eng production cost} = C_{\text{eng}} = 3112[0.043 T_{\max} + 243.25 M_{\max} + 0.969 T_{\text{turbine inlet}} - 2228] \quad \{\text{fps}\}$$
$$= 3112[9.66 T_{\max} + 243.25 M_{\max} + 1.74 T_{\text{turbine inlet}} - 2228] \quad \{\text{mks}\}$$
$$\text{RDT\&E} + \text{flyaway} = H_E R_E + H_T R_T + H_M R_M + H_Q R_Q + C_D + C_F + C_M + C_{\text{eng}} N_{\text{eng}} + C_{\text{avionics}}$$

DAPCA: Development and procurement costs of aircraft

(Works reasonably well for most aircraft types; 2012 dollars)

W_e = empty weight [(lb) or {kg}]
V = maximum velocity [(kt) or{km/h}]
Q = lesser of production quantity or number to be produced in five years

FTA = number of flight-test aircraft (typically 2–6)
N_{eng} = total production quantity times number of engines per aircraft
$T_{\max}$ = engine maximum thrust [(lb) or {kN}]
$M_{\max}$ = engine maximum Mach number
$T_{\text{turbine inlet}}$ = turbine inlet temperature [(°R) or {K}]
C_{avionics} = avionics cost

Engineering	$R_E = \$115$
Tooling	$R_T = \$118$
Quality control	$R_Q = \$108$
Manufacturing	$R_M = \$98$

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Time value of money

$$NPV = \frac{FV}{(1+r)^n}$$

NPV is the net present value

FV is the future value

r is the discount rate—potential return if invested

n is the number of years into the future

Example

The value today of a cash flow of $121 in two years at a 10% discount rate is $100:

$$NPV = \frac{FV}{(1+r)^n}$$

$$NPV = \frac{\$121}{(1+0.1)^2}$$

$$NPV = \$100$$

Learning curve effects

H : production cost (in labor hours) per aircraft

H_1 : initial production cost (in labor hours) per aircraft

Q : production quantity

x : learning curve, e.g., 80% means 20% lower cost each time the quantity doubles

$$H = H_1 Q^{\log_2(\text{learning curve})}$$
$$= H_1 Q^{\log(\%\ \text{learning curve}/100)/\log 2}$$

If H_1=10,000 hours and x=80%, then

Q=1 → H=10,000 hours

Q=2 → H=8,000 hours

Q=4 → H=6,400 hours

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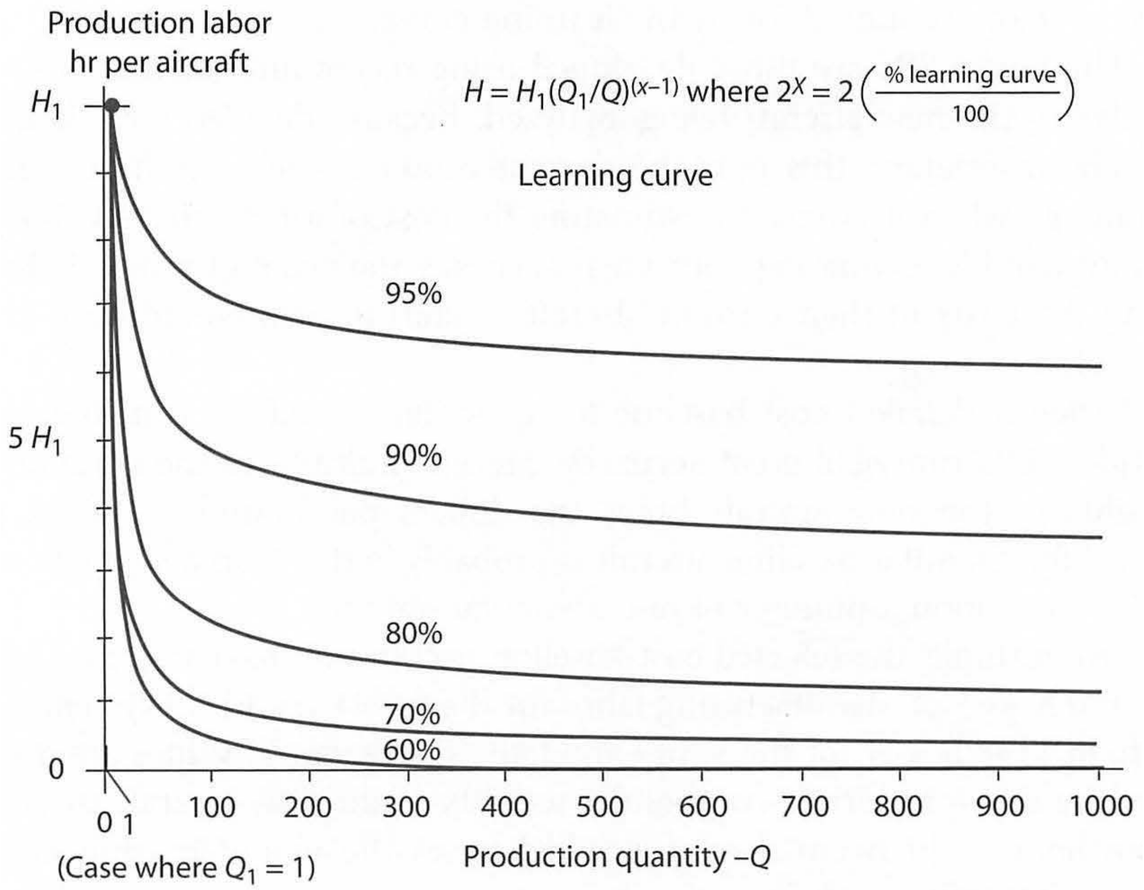

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Investment cost analysis—airline's perspective

- Bottom line: Is the net present value (NPV) of the total profit greater than the investment cost (purchase price)?
- Net present value (NPV): The value of something today (i.e., in 2022 dollars) based on an assumed rate of return. For example, something worth $110 next year has an NPV of $100 today assuming a 10% expected rate of return.
- In general, the value after n years is $FV = NPV(1 + r)^n$ where r is rate of return and NPV is the value today.
- The rate of return is also known as the discount factor, since it determines how much the NPV is lower than the actual amount of revenue or cost some time in the future.
- The NPV of a commercial aircraft is the sum of all the NPVs of the incoming cash flow minus the TOC (DOC + IOC). We do not include depreciation, since we consider the purchase price separately, but we do add the NPV of the salvage value (typically 10% of the purchase price).
- The value of r for which the NPV of total profit is exactly the purchase price is called the internal rate of return —i.e., the return of the investment of purchasing the aircraft.
- Operator's perspective for military aircraft: NPV of delivered capability versus acquisition and operation cost

Break-even analysis—manufacturer's perspective

- Break-even point: The time when enough aircraft have been sold that revenues have paid off development costs.
- Sales price: Based on market effects as well as the cost to develop the aircraft, certify it, and manufacture it.
- Contribution margin: Cost to produce an aircraft minus the sales price. This initially contributes to paying off the development costs and then earns profit for the manufacturer.
- Example: An aircraft costs $400 million to develop, and the contribution margin is $2 million (thus, the sales price is $2 million more than the production cost). Then the break-even point is 200 aircraft.

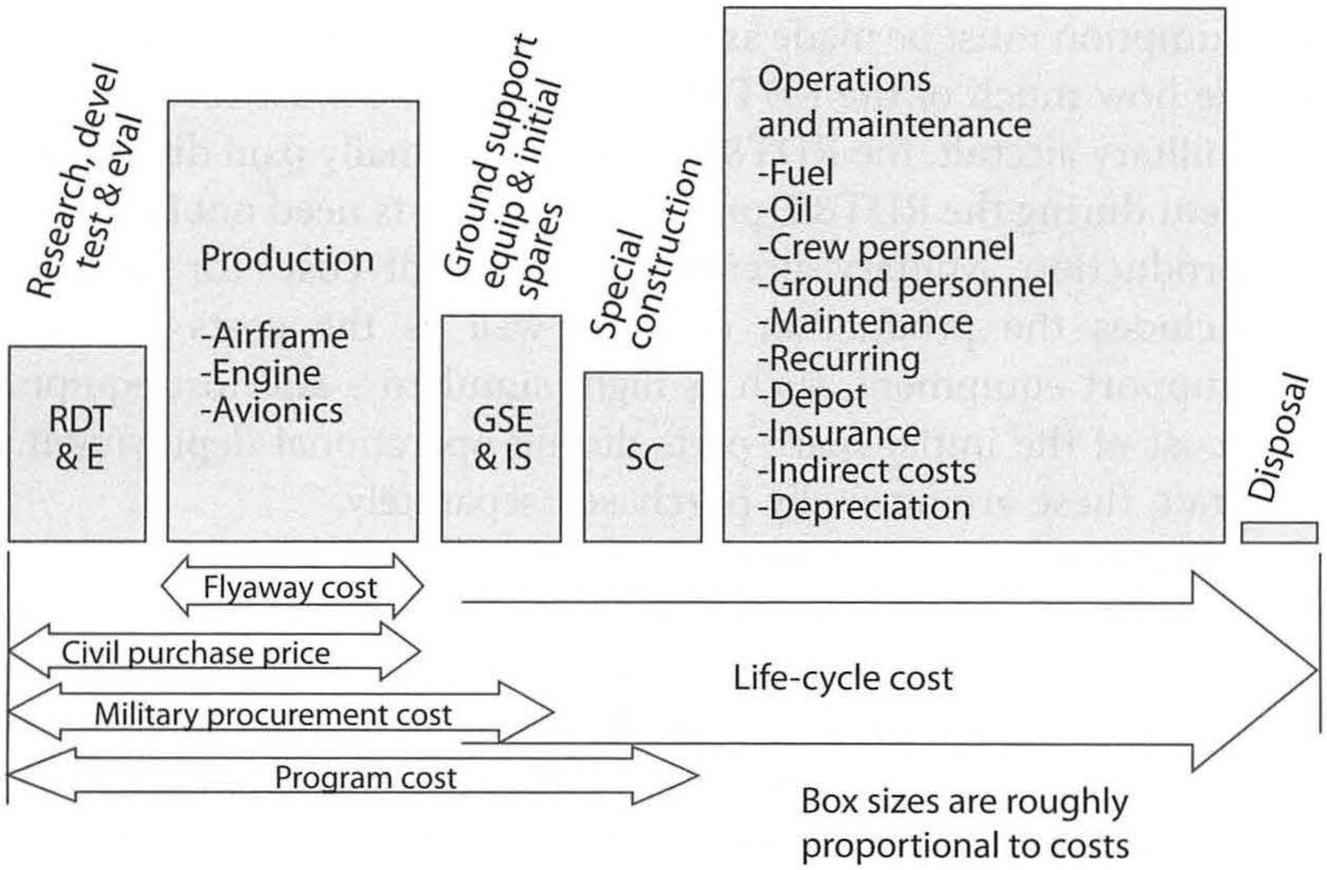
Summary

Commercial aircraft:

- DOC per seat mile (O&M)
- Investment cost analysis
- Break-even analysis

Military aircraft:

- Cost per flying hour (CPFH)
- Lifecycle cost



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